Dr. Raghuvir Singh Tomar**,** Professor
Department of Biological Sciences

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Area of research interest: Biochemistry and Molecular Biology; Understanding the epigenetic regulation of stress response pathways in yeast and poultry animals

Administrative experience:

Coordinator of Department of Biological Sciences, IISER Bhopal: 2010 - 2012

Head, Department of Biological sciences, IISER Bhopal: January 2015 - March 2018

Dean, Student affairs, IISER Bhopal: August 2018 - September 2022

Head, Center of Science and Society (CS2), IISER Bhopal, January 2023 onwards

Professional Experience

B.Sc.: in Biology from Jiwaji University, Gwalior, INDIA, 1991

M.Sc.: in Biochemistry from Jiwaji University, Gwalior, INDIA, 1993

Ph.D.: Ph.D. in Zoology, 2001 from Center for Advanced Study in Zoology, Banaras Hindu University, Varanasi, INDIA

Post Doctoral Fellow, May 2002-September 2003: at University of Texas, Southwestern Medical Centre, Dallas (USA).

Post Doctoral Fellow, October 2003 - December 2006, at The Penn State University, University Park (USA).

Post doctoral Fellow, January 2007- December 2008, at The Ohio state university, Columbus (USA).

Assistant Professor, Biological Sciences, January 2009 - November 2013: Indian institute of Science Education and Research Bhopal, India.

Associate Professor, Biological Sciences, December 2013 - February 2018: Indian institute of Science Education and Research Bhopal, India.

Professor, Biological Sciences, February 2018 onward: Indian institute of Science Education and Research Bhopal, India.

Academic/professional honours:

- Obtained 1st rank in the college during B.Sc. Based on merit, selected as secretary for student union.
- Junior Research Fellowship by UGC for Ph.D.

- Invited as member in task force meeting for evaluation of research projects by Department of Biotechnology, Govt. Of India
- Invited by several institutes: IISc, CCMB, CDRI, IICT, IIT Roorkee, NIT Allahabad, BHU, JNU, NIT Bhopal as external expert member to conduct Ph.D. thesis defence examinations.
- Invited by CSIR, DBT, CSIR and C-CAMP to evaluate various research projects.
- Reviewed manuscripts for journals; EMBO Reports, FEBS Journal, ACS Applied Materials & Interfaces, Oncotarget, Mol Biol Reports, Autophaby, Environmental Pollution, PlosOne, Communications Biology, Molecular and Cellular Biology etc.
- Served as member, board of governors, IISER Bhopal.

Courses taught:

Bio 101: Biomolecules and origin of life

Bio 202: Basic genetics

Bio 104: General Biology laboratory II

Bio 102: Diversity in the living world

Bio 301: Cell Biology

Bio 303: Biochemistry I

Bio 302: Biochemistry II

Bio 304: Molecular Biology

Bio 307: Biology laboratory I

Bio 308: Biology laboratory II

Bio 501S/601S: Epigenetics

Bio 201: Fundamentals of Molecular Biology

Bio 102: Fundamentals of Cell Biology

Ph.D. students registered at present:

1. Santoshi Acharjee, 2110106
2. Rajshree Pal, 2120112
3. Vandana Anjana, 2310114
4. Smriti Anand, 2320115
5. Prateeksha Thakur, 2320112
6. Sugandh Sharma, 2410125
7. Lakshmi Venkatesh Kolhapur, 2510112
8. Pooja Sharma, 2520110

Ph.D. students supervised:

1. Dr. Papita Mandal, Post-Doctoral Fellow at King Abdullah University Science Technology, Saudi Arabia.
2. Dr. Gajendra Kumar Azad, Assistant Professor, department of Zoology, University of Patna.
3. Dr. Vikash Singh, Post-doctoral fellow at Penn State University, USA.
4. Dr. Sakshi Chauhan, Post-doctoral fellow at NIH, USA
5. Dr. Upendarrao Golla, Post-doctoral fellow at Penn State University, USA.
6. Dr. Pushpendra Kumar Sahu, Post-doctoral fellow at University of California, San Francisco.
7. Dr. Pilendra Kumar Thakre, Post-doctoral fellow at Baylor College of Medicine, USA
8. Dr. Santhosh Kumar Sariki, Post-doctoral fellow at University of Michigan, USA
9. Dr. Swati Swagatika, Assistant Scientist in Pollution control board, Orissa
10. D. Rakesh Kumar Sahu, Post-doctoral fellow at NIH, USA
11. Dr. Sakshi Singh, Post-doctoral fellow at NIH, USA
12. Dr. Ramesh Kumawat, joined the laboratory of Prof. Jerry Workman at Stowers Institute for Medical Research
13. Nitu Saha, Post Doc at Yale school of medicine
14. Ranu Singh, 1820114, Ph.D. completed, now looking for job.

Students supervised for BS-MS thesis

1. Naveen Verma, 08012, BS-MS 2013
2. Goutham Bandi, 09014, BS-MS 2014
3. Jagriti Banerjee, 10032 BS-MS 2015
4. Rishith Ravindran, 10071, BS-MS 2015
5. Deepthi Joseph, 10024, BS-MS 2015
6. Mubthasima PP, 11045, BS-MS 2016
7. Jeena TJ, 11035, BS-MS 2016
8. Sagar Salim, 12093, BS-MS 2017
9. Athira SV, 12027, BS-MS 2017
10. Soumya Prakash Sahu, 11065, BS-MS 2017
11. Rohit Kumar, 13115, BS-MS, 2018
12. Mahajan Vibhuti Prakash, 13077, BS-MS 2018
13. Yagna Anand Oza, 14102, BS-MS 2019
14. Anaswara Sugathan, Roll No., 15194
15. Harsha P M, Roll No., 15072
16. Debashish Naik, Roll No. 16057
17. Manali Powar, Roll No. 16104

18. Srobona Mahajan, 17291

19. Ramakrishnan PS, 18194

20. Nayanika Mukherjee, 18154

21. Sahil Choudhary, 19258

22. Ritika Manjhi, 20228

23. Sharayu Subhash Awachat, 20255

24. Devika Rejidev, 20101

M.Sc. students supervised

1. Mrittika Paul, 220103

Research Projects: Completed, ongoing and submitted

S. No.	Title of Projects	Funding Agency	Duration	Status
1.	Biochemical characterization of Hat1	DBT	2010 - 2013	Completed
2.	Epigenetic regulation of Aging	DBT	2012 - 2015	Completed
3.	Identification of cellular and molecular mechanism behind the action of Ebselen	CSIR	2011 - 2014	Completed
4.	Mechanism of transcription and chromatin remodelling: Role of Repressor Activator Protein 1 (Rap1) in regulation of DNA repair genes	DST	2013 - 2016	Completed
5.	To study role of yeast Sen1 in regulation of Lipid and Redox Homeostasis	SERB	2016 - 2019	Completed
6.	To dissect the molecular mechanism of flocculation and biofilm formation of yeast.	CSIR	2018 - 2021	Ongoing
7.	To characterize the epigenetic target of potential anti-cancer drug: Curcumin, Ebselen and KP1019	CSIR	2018 - 2021	Ongoing

Conferences/meetings attended:

1. Indo-US Science Symposium organized by Indo-US Science and Technology Forum, held at Jaypee Palace hotel, Agra from March 1 – 4, 2009. Poster presented.

2. Young Investigator meeting held at Mayfair hotel, Bhubaneswar, India from February 13 -16, 2011. Poster presented.

3. Attended a conference ‘Chromatin and Epigenetics, from 1–5 June 2011, Heidelberg, organized by EMBL Advanced Training centre, Heidelberg, Germany. Poster title: N-terminal clipping of Histone H3 regulates Chaperone function of NASP.

4. Invited talk at C-CAMP, NCBS in Technology Conclave, November 14, 2014.

5. Invited speaker in INSA meeting hosted by IISER Bhopal, 28 – 30 December, 2015. Title of talk: Yeast Sen1 is essential for maintenance of cellular redox homeostasis and life span.

6. Invited speaker, Title of talk: Novel function of Yeast Sen1 in genomic and cell wall integrity, 5th Chromatin Asia Meeting held at JNCSR, Bangalore, January 14 – 19, 2015.

7. Invited speaker in in 26th National Congress of Parasitology at Department of Biochemistry, Banaras Hindu University, Varanasi, Dec 21 – 24th, 2015. Title of talk was ‘Yeast Crg1 regulates cell cycle by modulating Target of Rapamycin (TOR) pathway and mitochondria’

8. Represented IISER Bhopal in YIM2016 held at Heritage Village Resort & Spa, Manesar from Feb 27 – 28, 2016.

9. Invited speaker in “National Symposium on Current Research in Cancer Biology & Therapy” organized by School of Biological Science & Biotechnology, University & Institute of Advanced Research, Gandhinagar, October 7-8, 2016". The title of talk ‘Clipping of histone tails is an epigenetic event that regulates structure and function of chromatin.

10. Delivered a talk entitled ‘Genetic and epigenetic mechanisms maintain cellular homeostasis by regulating gene expression’ at department of biochemistry, BHU, July 19, 2018.

11. Delivered an invited seminar entitled ‘The CWI MAP kinase pathway plays a critical role to regulate the expression of cell wall flocculins and Flocculation phenotype of yeast required for survival under stress conditions’ in ICERB-2018 conference organized at Bilaspur University (CG) from October 28 – 30, 2018.

12. Delivered an invited talk entitled ‘Transcription of yeast FLO genes is regulated by Cell Wall Integrity pathway mediated antagonism between Rlm1 and Tup1-Ssn6 in conference ‘7th Asian forum of chromosome and chromatin biology’ organized by JNCASR, Bangalore from November 15th – 17th, 2018.

13. Delivered an invited lecture on a topic entitled ‘Dissecting the role of epigenetics in regulation of cellular homeostatic pathways under stress conditions’ in the 2nd International Conference on Recent Advances in Biotechnology and Nanobiotechnology (Int-BIONANO-2022) during 10-11 February 2022 organized by the Amity University, Gwalior.

14. Delivered an invited lecture on a topic entitled ‘Copper homeostasis is tightly regulated; excess Copper can inhibit the Sec61 protein translocation channel’ in the 12th International Conference on Yeast Biology (Yeast India 2023) during 10-13th March, 2023 organized by IISER Mohali.

15. Invited speaker, Title of talk: Copper homeostasis is epigenetically regulated; excess Copper inhibit the protein translocation, 8th Chromatin Asia Meeting held at JNCSR, Bangalore, November 4 – 6, 2023.

16. Invited speaker, International Conference on Metabolism, Toxicology, Diagnostics and Therapeutics, 28-30th January 2026, organized by Department of Biomedical and Life Sciences, School of Science, Navrachana University, Vadodara. The title of talk was 'Epigenetics as a regulatory layer connecting the metallome and metabolome'.

Awards of Ph.D. scholars:

1. Papita Mandal got selected for prestigious **'**Keystone Symposia scholarship' funded by the German Ministry of Education and Research to present a poster in the Keystone symposia meeting 'Chromatin Mechanisms and Cell Physiology**'** at Oberstdorf, Germany, March 23 – 28, 2014
2. Travel award to Gajendra K. Azad from Department of Science and Technology, Govt. of India for presenting a poster in Wellcome Trust meeting **'**Chromatin: from nucleosomes to chromosomes' at Wellcome Trust Genome Campus, Hinxton, Cambridge, UK, April 30 – May 2, 2014
3. Travel award to Vikash Singh from DST to present his work in 'Chromatin and Transcription' meeting held at EMBL, Heidelberg, Germany from 23 - 26 August, 2014.
4. Travel award by DST to Upendarrao Golla for presenting two posters in 27th International conference on Yeast Genetics and Molecular Biology (ICYGMB), Levico Terme, Trento, Italy. Sept 06-12, 2015.
5. Sakshi Chauhan presented a poster entitled 'Characterization of H3 tail clipping activity of H3 protease Glutamate dehydrogenase' in a conference 'Gene Networks in Chromatin/Chromosome functions', 5th Chromatin Asia international meeting held at JNCSR, Bangalore, January 15th – 18th, 2015.
6. Travel award by DST to Sakshi Chauhan to present a poster in Keystone Symposium on "Epigenetics and Human Disease: Progress from Mechanisms to Therapeutics, January 29 - February 2nd, 2017, organized at Sheraton Seattle hotel, Seattle, Washington, USA. Poster title: Clipping of histone H3 is regulated by multiple factors; structure and post-translational modifications of histone H3 and stefin B.
7. Pushpendra Kumar Sahu selected for best poster award inXL All India Cell Biology Conference & International Symposium on Functional Genomics and Epigenomics, organized by Jiwaji University, Gwalior from November 17-19, 2016. The title of the poster "Yeast Crg1 is required for mitochondrial integrity and TOR signalling by modulating redox homeostasis*"*.
8. Santhosh Kumar Sariki selected for best poster award in the 11th International Conference on "Biology of Yeast and Filamentous Fungi" held at the School of Life Sciences, University of Hyderabad, Hyderabad during November 27-29, 2019.
9. Travel award by DBT and SERB to Ranu Singh. Presented a poster in Gordon research conference "histone and DNA modifications" held at Lucca (Burga), Italy from July 20 - 25, 2025.
10. Ranu Singh presented a poster and received the best flash talk award in the 13th International Conference of Yeast Biology held at IISc Bangalore, scheduled from December 3 to 6, 2025.

Research interests:

To identify the role of epigenetics in maintenance of cellular homeostatic pathways and stress:

In eukaryotes, DNA is organized in the form of chromatin which consists of repeating units called nucleosomes. The structure and function of chromatin is regulated by the specific enzymes that modify DNA and histone proteins to maintain cellular homeostatic mechanisms under a variety of pathophysiological conditions. Several classes of enzymes compete for binding to the chromatin. The coordination among chromatin modifying enzymes is critical for survival under various stress conditions. The imbalance in the activity of chromatin modifying enzymes leads to the development of disease pathologies. We trying to understand the significance of chromatin associated epigenetic changes in response to intra and extracellular factors. Following are our key research questions:

1. To identify the role of histone tails in regulation of yeast flocculation.
2. To investigate the significance if histone clipping in viral and bacterial infections.
3. To identify the factors regulating the cellular homeostasis of metal ions.
4. To understand the significance of lipid droplets in DNA damage response and metal homeostasis.

List of publications:

1. **Tomar R. S.**, Psathas J. N., Zhang J. N., Zhang Z. and Reese J. C. A novel mechanism of antagonism between chromatin remodeling complexes regulates RNR3 expression. **Molecular and Cellular Biol**ogy, 29, 3255-3265, 2009.
2. **Tomar R. S.**, Zheng S., Reese D. B., Walcott H. N. and Joseph C. Reese. Yeast Rap1 contributes to genomic integrity by activating DNA damage repair genes. **EMBO Journal**, 27, 1575-1584, 2008.
3. Sharma V. M., **Tomar R. S.**, Dempsey A. E., and Reese J. C. Histone deacetylases RPD3 and HOS2 regulate transcriptional activation of DNA damage inducible genes. **Molecular and Cellular Biol**ogy, 27, 3199-3210, 2007.
4. **Tomar R. S.**, Matta H. and Chaudhary P. M. Use of adeno associated viral vectors for delivery of Small interfering RNA, **Oncogene**, 22, 5712-5715, 2003.
5. Kauppila S., Walid S. A. Maaty Waliod S. A., Chen P., **Tomar R. S.**, Michael T. Eby, Joe Chapo, Sukit Chew, Nisha Rathore, Sunny Zachariah, John M. Abrams and Preet M. Chaudhary. Eiger and its receptor, Wengen, comprise a TNF-like system in Drosophila. **Oncogene**, 22, 4860-4867, 2003.
6. Panigrahi AK., **Tomar R. S.** and Chaturvedi M. M. Mechanism of nucleosome disruption and octamer transfer by the chicken SWI/SNF -like complex. **BBRC,** 306, 72-78, 2003.

7. Panigrahi AK., **Tomar RS.** and Chaturvedi MM. A SWI/SNF-like Factor from Chicken Liver that Disrupts Nucleosomes and Transfers Histone Octamers in cis and trans. **Archives of Biochemistry and Biophysics**, 414: 24-33, 2003.
8. Matta H.*, Hozayev B.*, **Tomar R. S.***, Chugh P., and Chaudhary P. M. Use of Lentiviral Vectors for Delivery of Small Interfering RNA. **Cancer Biology and Therapy**, 2, 206-210, 2003.
9. Mandal P, Azad GK & **Tomar, R. S.** (2012) Identification of a novel histone H3 specific protease activity in nuclei of chicken liver. **BBRC**, 421: 261- 267.
10. Azad GK, Balkrishna SJ, Sathish N, Kumar S & Tomar RS* (2012) Multifunctional Ebselen drug functions through the activation of DNA damage response and alterations in nuclear proteins. **Biochemical Pharmacology,** 83: 296-303.
11. Mandal P, Verma N, Chauhan S & Tomar RS* (2013) Unexpected histone H3 tail clipping activity of Glutamate dehydrogenase**. Journal of Biological Chem**istry, 288: 18743-18757.
12. Golla U, Singh V, Azad GK, Singh P, Verma N, Mandal P, Chauhan S & Tomar RS* (2013) Sen1p contributes to genomic integrity by regulating expression of Ribonucleotide reductase 1 (RNR1) in Saccharomyces cerevisiae. **PLoS One** 8: e64798.
13. Azad GK, Singh V, Golla U & Tomar, RS* (2013) Depletion of cellular iron by curcumin leads to alteration in histone acetylation and degradation of Sml1p in Saccharomyces cerevisiae. **PLoS One** 8: e59003.
14. Mandal P, Verma N, Azad GK, Singh V, Golla U & Tomar RS (2014) Epigenetics: Role of histone proteases in cellular functions and diseases. **A book chapter** in 'Molecular Mechanisms and Physiology of Disease: Implications for Epigenetics and Health'. Maulik, N.; Karagiannis, T. (Eds.), Springer 2014.
15. Azad GK & Tomar RS* (2014) Proteolytic clipping of histone tails: The emerging role of histone proteases in regulation of various biological processes. **Mol Biol Rep** 41: 2717-2730.
16. Azad GK & Tomar RS* (2014) A comprehensive review illustrating the mode of action and biological pathways targeted by promising antioxidant drug, ebselen. **Mol Biol Rep** 41: 4865 – 4879.
17. Azad GK, Singh V, Mandal P, Singh P, Golla U, Baranwal S, Chauhan S & Tomar RS* (2014) Ebselen induces reactive oxygen species (ROS)-mediated cytotoxicity in Saccharomyces cerevisiae with inhibition of glutamate dehydrogenase being a target. **FEBS Open Biol** 4: 77 – 89.
18. Balkrishna SJ, Kumar S, Azad GK, Bhakuni BS, Panini P, Ahalawat N, Tomar RS, Detty MR & Kumar S* (2014) An Ebselen like Catalyst with Enhanced GPx Activity via a Selenol Intermediate. **Org Biomol Chem** 12: 1215 – 19.
19. Singh V, Azad GK, Mandal P, Reddy AM & Tomar RS* (2014) Anti-cancer drug KP1019 modulates epigenetics and induces DNA damage response in Saccharomyces cerevisiae. **FEBS Letters,** 588: 1044 - 1052.

20. Singh V, Azad GK, Mandal P, Reddy A.M, Baranwal S & Tomar RS* (2014) Anti-cancer drug KP1019 induces Hog1 phosphorylation and protein ubiquitination in Saccharomyces cerevisiae. **European Journal of Pharmacology**,736: 77 – 85.
21. Azad GK, Singh V & Tomar RS* (2014) Assessment of the biological pathways targeted by isocyanate using N-succinimidyl N-methylcarbamate on budding yeast Saccharomyces cerevisiae. **PLoS One 9: e92993**.
22. Baranwal S, Azad GK, Singh V & Tomar RS* (2014) Signaling of chloroquine-induced stress in the yeast Saccharomyces cerevisiae requires the Hog1 and Slt2 MAPK Pathways. **Antimicrobial Agents Chemotherapy,** 58: 5552 – 5566.
23. Mandal P, Chauhan S & Tomar, RS* (2014) H3 clipping activity of glutamate dehydrogenase is regulated by stefin B and chromatin structure. **FEBS Journal,** 281: 5292 – 5308.
24. Azad GK, Singh V, Thakare MJ, Baranwal S & Tomar RS* (2014) Mitogen-activated protein kinase Hog1 is activated in response to curcumin exposure in the budding yeast Saccharomyces cerevisiae. **BMC Microbiology,** 14: 317-327.
25. Azad GK, Singh V, Baranwal S, Thakare MJ & Tomar RS* (2015) The transcription factor Rap1p is required for tolerance to cell wall-perturbing agents and for cell wall maintenance in Saccharomyces cerevisiae. **FEBS Letters,** 589: 59 – 69.
26. Golla U, Goutham B & Tomar RS* (2015) Molecular Cytotoxicity Mechanisms of the Herbicide Allyl Alcohol (Acrolein) in Budding Yeast. **Chemical Research in Toxicology,** 28: 1246 – 1264.
27. Singh V, Azad GK, Sariki SK & Tomar RS (2015) Flocculation in Saccharomyces cerevisiae is regulated by RNA/DNA helicase Sen1p. **FEBS Letters,** 589: 3165 – 3174.
28. Chauhan S and Tomar RS* (2016) Efficient expression and purification of biologically active human cystatin proteins. **Protein Expression and Purification,** 118: 10 – 17.
29. Azad GK and Tomar RS (2016) Partial purification of histone H3 proteolytic activity from budding yeast Saccharomyces cerevisiae. **Yeast**, 33, 217-226.
30. Azad GK, Singh and Tomar RS* (2016) The multifunctional transcription factor Rap1: a regulator of yeast physiology. **Frontiers Bioscience**, 21, 918 – 930.
31. Golla U, Adhikary A, Mondal AK, Tomar RS* and Konar S* (2016). Synthesis, structure, magnetic and biological activity studies of bis-hydrazone derived Cu(II) and Co(II) coordination compounds. **Dalton Trans**, 55, 11849 – 11863.
32. Chauhan S, Mandal P and Tomar RS* (2016) Biochemical Analysis Reveals the Multifactorial Mechanism of Histone H3 Clipping by Chicken Liver Histone H3 Protease. **Biochemistry**, 55: 5464 – 5482.
33. Golla U, Joseph D and Tomar RS* (2016). Combined Transcriptomics and Chemical-Genetics Reveal Molecular Mode of Action of Valproic acid, an Anticancer Molecule using Budding Yeast Model". **Scientific Reports**, 35322: doi:10.1038/srep35322.

34. Sariki SK, Sahu PK, Golla U, Singh V, Azad GK and Tomar RS* (2016). Sen1, the homolog of human Senataxin, is critical for survival through regulation of cellular redox homeostasis, mitochondrial function and TOR in Saccharomyces cerevisiae. **FEBS Journal**: 283, 4056 - 4083.
35. Chauhan S, Azad G K and Tomar R S. *In vitro* histone H3 cleavage assay for yeast and chicken liver H3 Protease. (2017) ***Bioprotocol***7 10.21769/BioProtoc.2085.
36. Swagatika S. and Tomar RS. **A book chapter**, title: Modulation of Epigenetics by Environmental Toxic Molecules in 'Advances in Molecular Toxicology, vol.: 10, 2016, Article Number: 8, page: 1 – 29, published by Elsevier.
37. Sekhar AR, Sariki SK, Reddy RV, Bisai A, Sahu PK, Tomar RS, Sankar J* (2017). Zwitterionic BODIPYs with large stokes shift: small molecular biomarkers for live cells. **Chem Commun**. 53: 1096 – 1099.
38. Sk B#, Thakre P#, Tomar RS*, Patra A* (2017). A pyridoindole based multifunctional bioprobe: pH-induced fluorescence switching and specific targeting of lipid droplets. (#equal contributor, *equal corresponding author). **Chem. Asian J**., 12, 2501 – 2509.
39. Golla U, Swagatika S, Chauhan S and Tomar RS* (2017). A Systematic Assessment of Chemical, Genetic, and Epigenetic Factors Influencing the Activity of Anticancer Drug KP1019 (FFC14A). **Oncotarget**, 8, 98426 – 98454.
40. Saha N and Tomar RS* (2018. **A book chapter**, title: 'Epigenetics and Infectious Pathogens: Interactions, Ploy and Perspectives' chapter 20 in Infectious Diseases and Your Health, ISBN; 978-981-13-1576-3. Published by Springer Nature.
41. Azad GK, Swagatika S, Kumawat M, Kumawat R and Tomar RS* (2018). Modifying Chromatin by Histone Tail Clipping. **Journal of Molecular Biology**, 430, 3051-3067.
42. Babele PK, Thakre PK, Kumawat R and Tomar RS* (2018). ZnO-nanoparticles induce toxicity by affecting Cell Wall Integrity pathway, mitochondrial function and lipid homeostasis in *Saccharomyces cerevisiae.* **Chemosphere**, 213, 65-75.
43. Thakre P, Athira SV, Golla U, Chauhan S and Tomar RS* (2018). Previously uncharacterized amino acid residues in histone H3 and H4 mutants with roles in DNA damage repair response and cellular aging. **FEBS Journal**, 286, 1154-1173.
44. Sahu PK and Tomar RS* (2019). The natural anticancer agent cantharidin alters GPI-anchored protein sorting by targeting Cdc1-mediated remodeling in endoplasmic reticulum. **Journal of Biological Chemistry**, 294, 3837-3852.
45. Sahu PK, Chauhan S and Tomar RS* (2019). Crg1 N-terminus is essential for methyltransferase activity and Cantharidin resistance in *Saccharomyces cerevisiae*. **Biochemistry**, 58, 1799-1809.

46. Kumar P, Swagatika S, Dasari S, Tomar RS* and Patra AK* (2019). Modulation of ruthenium anticancer drugs analogs with tolfenamic acid: Reactivity, biological interactions and growth inhibition of yeast cell. **Journal Inorg. Biochem.**, 199, 110769. * equal corresponding.
47. Sahu PK, Salim S, Pp M, Chauhan S and **Tomar R. S.** (2019). Reverse genetic analysis of yeast YPR099C/MRPL51 reveals a critical role of both overlapping ORFs in respiratory growth and MRPL51 in mitochondrial DNA maintenance. **FEMS Yeast Research,** 19, foz056. doi.org/10.1093/femsyr/foz056
48. Sariki SK, Kumawat R, Singh V and Tomar RS* (2019). Flocculation of Saccharomyces cerevisiae is dependent on activation of Slt2 and Rlm1 regulated by the Cell Wall Integrity pathway. **Molecular Microbiology,** 112,1350-1369.
49. Sahu R, Saha N, Das L, Sahu PK, Sariki SK and **Tomar R. S.** (2020). SWI/SNF chromatin remodeling complex contributes to clearance of cytoplasmic protein aggregates and maintains basal unfolded protein response in Saccharomyces cerevisiae. **FEBS Journal**, 287, 3024-3041.
50. Thakre PK, Golla U, Biswas A, and **Tomar R. S.** (2020). Identification of histone H3 and H4 amino acid residues essential for regulation of arsenite stress signaling in Saccharomyces cerevisiae. **Chemical Research in Toxicology,** 33, 817-833**.**
51. Sahu RK, Singh S, and Tomar RS (2020). The mechanisms of action of chromatin remodelers and implications in development and disease. **Biochemical Pharmacol**ogy, doi: 10.1016/j.bcp.2020.114200.
52. Singh S, Sahu RK and Tomar RS (2020). The N-terminal Tail of Histone H3 Regulates Copper Homeostasis in *Saccharomyces cerevisiae*. **Molecular and Cellular Biology**, doi: 10.1128/MCB.00210-20.
53. Swagatika S. and Tomar RS (2021). ABC transporter Pdr5 is required for cantharidin resistance in Saccharomyces cerevisiae. **BBRC,** doi: 10.1016/j.bbrc.2021.03.074.
54. Swagatika S. and Tomar RS (2021) Cantharidin downregulates PSD1 expression and inhibits autophagic flux in yeast cells. **FEBS Open Bio.,** doi: 10.1002/2211-5463.13196.
55. Thakre PK, Sahu RK, and Tomar RS (2021). Substitution of histone H3 arginine 72 to alanine leads to deregulation of isoleucine biosynthesis in budding yeast Saccharomyces cerevisiae. **Biochem Cell Biol**., doi: 10.1139/bcb-2020-0651
56. Sahu RK, Singh S and Tomar RS (2021). The ATP-dependent SWI/SNF and RSC chromatin remodelers cooperatively induce unfolded protein response genes during endoplasmic reticulum stress. **BBA-Gene Regulatory Mechanisms**, doi.org/10.1016/j.bbagrm.2021.194748
57. Singh S, Sahu RK, Sugathan A and Tomar RS (2021). The H2A N-terminal tail is required to alleviate copper-induced stress in Saccharomyces cerevisiae. **FEMS Yeast Research**, doi.org/10.1093/femsyr/foab061
58. Kumawat R. and Tomar R. S (2022). Heavy metal exposure induces Yap1 and Hac1 mediated derepression of GSH1 and KAR2 by Tup1-Cyc8 complex. **Journal of Hazardous Materials,** 429, 128367. doi.org/10.1016/j.jhazmat.2022.128367

59. Saha N and Tomar R S (2022). Copper inhibits protein maturation in the secretory pathway by targeting the Sec61 translocon in Saccharomyces cerevisiae. Accepted for publication in **Journal of Biological Chem**istry, doi.org/10.1016/j.jbc.2022.102170
60. Acharjee S, Chauhan S, Pal R and Tomar RS (2023). Book chapter 'Mechanisms of DNA methylation and histone modifications' in a Elsevier press book 'Progress in Molecular Biology and Translational Science, 197, 51-92
61. Kundu S, Bahadur Sk, Saha N, Das S, Dutta T K, Batra A, Tomar R S* and Patra A* (2023) Unraveling Molecular Assembly and Tracking Lipid Droplet Dynamics Using Fluorescent Phenanthroimidazole Derivatives. **ACS Materials Lett**., 5, 27−35. *equal corresponding.
62. Saha N, Swagatika S and Tomar R S (2023). Investigation of the acetic acid stress response in Saccharomyces cerevisiae with mutated H3 residues. **Microbial Cell, *10: 217-232.* doi: 10.15698/mic2023.10.806.**
63. Saha N, Acharjee S and Tomar R S (2024). Cdc73 is a major regulator of apoptosis-inducing factor 1 expression in Saccharomyces cerevisiae via H3K36 methylation. **FEBS Lett:** 598, 658-669.
64. Chauhan S and Tomar RS (2024). Unveiling the molecular networks underlying cellular impairment in Saccharomyces cerevisiae: investigating the effects of magnesium oxide nanoparticles on cell wall integrity and endoplasmic reticulum stress response. **Environ Sci Pollut Res Int**: 31, 30149 – 30162.
65. Kumawat R. and Tomar R. S (2024). Dissecting the role of HOG mitogen-activated protein kinase signaling in the regulation of Yeast Flocculation. **FEBS Journal**. DOI: https://doi.org/10.1111/febs.17137.
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